

Reconstructed Time

For each hit, the coarse and fine timestamp is converted to ns as:

$$\begin{aligned} T_{\text{tick}} &= 32768 \text{ rollover} + \text{coarse}, & \Delta t_{\text{clk}} &= \frac{1000}{\text{clock_MHz}}, \\ i_{\text{TDC}} &= \text{tdc} + 4 \text{ pixel} + 16 \text{ column} + 128 \text{ fifo}, & t_0 &= (T_{\text{tick}} - \phi(\text{fine}, i_{\text{TDC}})) \Delta t_{\text{clk}}. \end{aligned}$$

ϕ is the calibrated fine-time fraction. It is read from the `hFineLut` LUT; if the LUT is disabled or missing, the code falls back to the linear min/max map for that TDC. With `-no-fine`, $\phi = 0$.

The TDC/channel-offset calibration applied while reading hits is:

$$t = t_0 - O_{\text{ch,tdc}}.$$

$O_{\text{ch,tdc}}$ is the relative channel/TDC offset stored in `hChannelTdcOffset`. By default it is extracted relative to channel 22; alternatively, an event-median reference can be used.

The ToT is computed within the same spill and channel by pairing each leading TDC with its trailing partner:

$$\text{ToT} = t_{\text{trailing},0} - t_{\text{leading},0}, \quad 0 < \text{ToT} \leq \text{-duration}.$$

The ToT uses the fine-calibrated raw time t_0 , not the time already corrected for offsets.

TDC and Offset Calibration

Basic command:

```
./script/run_tdc_calibration.sh -i ../data/RUN -d 50
```

It produces `calibration/TDC_calibration.root` and `output/TDC_calibration.pdf`. The script checks from the logbook that the run is in opmode 1 unless `-skip-logbook-check` is passed.

Main options:

- `-q-low`, `-q-high`, `-min-entries`: quantiles and minimum statistics for the fine LUT.
- `-match-coincidence` and `-d NS`: build the LUT using leading hits with valid ToT within the requested duration; default: use all leading hits.
- `-offset-study/-no-offset-study`: enable or disable the offset study, enabled by default.
- `-offset-reference 22`: default, reference offsets to channel 22.
- `-offset-reference event-median -offset-min-channels N`: use the event median as reference.
- `-offset-window NS` and `-offset-event-window NS`: windows used to associate hits from the same laser event.
- `-offset-channels logbook|all|CSV`: channels included in the offset study; default `logbook`.

TimeWalk Correction

The calibration uses the relation between sensor ToT and:

$$\Delta t = t_{\text{sensor}} - t_{\text{ref}}.$$

By default t_{ref} is the cleaned channel-22 laser trigger; `-reference-mode event-median` is available for event-reference studies. The fit gives a function $f_{\text{ch}}(\text{ToT})$, and the applied correction is:

$$t_{\text{TW}} = t - f_{\text{ch}}(\text{ToT}).$$

This aligns the corrected sensor time to the chosen reference. The default model is now **inverse-power**:

$$f(\text{ToT}) = p_0 + \frac{p_1}{(\text{ToT} - p_2)^{p_3}}.$$

Here p_0 is the asymptotic plateau, p_1 is the amplitude, p_2 is an effective threshold shift, and p_3 is a free exponent. The macro fills a raw TH2D Δt vs ToT, builds a TProfile along ToT, and fits the TProfile. If the free-exponent fit fails or has poor covariance quality, it retries with $p_3 = 1$. The older `pol1`, `pol1-plateau`, and `lin-exp-plateau` models remain selectable for comparison.

Basic command:

```
./script/run_timewalk_calibration.sh -i ../data/RUN -d 50 --signed-dt
```

It produces `calibration/timewalk_correction.root`. The macro applies the correction for closure plots and writes the `timewalk_corr_*` parameters. The standard analysis automatically uses this file when present; disable it with:

```
./script/run_analysis.sh -i ../data/RUN --no-timewalk
```

Available Cuts and Selections

- Active channels: read from the logbook; override with `-channels CSV` or `-sensors CSV`. In trigger mode, `-trigger 22` is removed from the sensor list.
- Fine wrap: `-fine-cut N` rejects hits with fine close to the TDC cut; `-no-fine` disables fine correction; `-no-lut` uses the linear min/max map.
- Valid ToT: required by default in timewalk mode; `-no-require-tot` allows leading hits without valid ToT. `-d NS` sets the maximum accepted ToT; `run_analysis.sh` also supports `-min-duration NS`.
- Spill range: `-spill-range MIN:MAX` reads only the selected inclusive spill range.
- Channel-22 trigger: `-trigger-tot-window MIN:MAX` cuts on trigger ToT; `-trigger-veto NS` rejects rebounds within NS after an accepted trigger; `-trigger-period NS` and `-trigger-period-tol NS` keep only triggers compatible with the expected period.
- Event/matching window: `-window NS` is the coincidence window; `-event-window NS` defines event building and defaults to `-window`; `-signed-dt` uses the nearest trigger and keeps the sign.
- TDC selection: `-tdc-selection 0` or `-tdc-selection 17:0,19:2,22:0`; the leading TDC is selected while the trailing partner is kept for ToT.
- Per-channel ToT window: `-channel-tot-window CH:MIN:MAX[,CH:MIN:MAX]`.
- Diagonal Δt /ToT cut: `-dt-tot-cut CH:DT0:SLOPE[:TOT_MIN:TOT_MAX]`. With the default `-dt-tot-cut-direction below`, the accepted region is $\Delta t \leq DT0 + SLOPE \cdot \text{ToT}$; above keeps the opposite side.
- Timewalk fit: `-timewalk-fit-ranges CH:MIN:MAX[,CH:MIN:MAX]` limits the TProfile points used by the fit; default `17:10:30,19:10:30`. `-timewalk-fit-model` selects `inverse-power`, `pol1`, `pol1-plateau`, or `lin-exp-plateau`.
- Final analysis: `run_analysis.sh` applies TDC/offset calibration with `-k calibration/TDC_calibration.root`; it applies timewalk with `-timewalk-calib FILE`, or automatically if `calibration/timewalk_correction.root` exists. `-window`, `-d`, `-min-duration`, and `-fine-cut` remain operational overrides.

Minimal Diagnostics

In the timewalk PDF, TH2 maps and TProfiles are printed on separate pages: use TH2 maps to inspect populations and cuts, and TProfiles to inspect the fit. The ROOT output stores the raw TH2D, TProfile, TF1 fit, corrected TH2D, and Δt projections before/after correction. The coincidence peak for the first sensor pair, normally 17–19, is saved before and after timewalk (`h_dt_uncorr_*`, `h_dt_corr_*`) and overlaid in a THStack in the PDF.